

Polymarket Trading Bot — Project Summary

Report

Sealed edition · Project span 2026-04-08 → 2026-09-20 (~5.5 months, incl. prehistoric v1) · Sealed 2026-09-20 · Archived together with all code and data

1. What this project is (one-sentence version)

A trading bot that uses AI (first Claude via manual prompt-pasting, later Zhipu GLM fully automated) to discover, buy, monitor and exit positions on Polymarket prediction markets. It evolved from "human decides, bot only executes take-profit/stop-loss" semi-automation all the way to a fully autonomous scan→discover→buy→re-evaluate→exit pipeline, finally running three live-money accounts as a controlled experiment on three exit strategies, and was sealed on 2026-09-20.

Four stages (incl. prehistory) + one side line:

① **Prehistoric v1 "polymarket-bot"** (2026-04-08 ~ 04-16): the earliest attempt — browser automation driving Gemini Deep Research for discovery; v2 survives only as a technical document.

① **v3 semi-auto initial** (2026-04-27, finalized and frozen within a day): human places orders; the bot only runs tiered take-profit/stop-loss.

② **Semi-auto main project v4 → v7.5.4** (2026-05-09 ~ 07-24): human pastes prompts to Claude and makes the calls, bot executes; 132 substantive commits; the stop-loss / take-profit / re-evaluation system matured here.

③ **Fully-automatic AUTO v8.0 → v8.5.5** (2026-07-06 fork ~ 2026-09-01 API stop): zero-human pipeline, GLM takes over all analysis; later expanded into the three-account experiment.

✚ **Side line: Weather bot** (since 2026-07-12, own wallet) — deterministic arbitrage on Polymarket daily-max-temperature markets (after sunset the temperature only falls; buy the 0.90–0.995 bucket for a 1–5% edge), fully isolated from the main line.

2. Final results (measured on seal day)

Per the project's iron rule the basis is **asset truth** (current total assets — net deposits); the per-trade ledger is detail only — worthless unsellable positions never enter the ledger, so the ledger

alone overstates performance. Numbers come from the read-only three-account report re-run on seal day, 2026-09-20.

Account / strategy	Net deposit	Assets at seal	P&L	Exit rule
Main · full auto strategy Kelly sizing + 3-tier SL + TP + GLM re-eval	\$99.66	\$90.61	—\$9.05 (—9.1%)	Full strategy: trailing/hard stops, 2x take-profit, intraday + daily re-evaluation
Bench · baseline takes every GLM recommendation	\$90.47	\$41.81	—\$48.66 (—53.8%)	One rule only: sell at —30%; otherwise hold to resolution
Shadow · low-price hold only buys recommendations ≤ 0.40	\$50.00	\$22.16	—\$27.84 (—55.7%)	—60% stop + \$0.05 floor; no take-profit, hold to resolution
Total	\$240.13	\$154.58	—\$85.55 (—35.6%)	

The most valuable findings of the whole experiment (from the three-account control):

- 1 Exit rules are everything.** On the same batch of GLM recommendations, the main account with full take-profit/stop-loss lost only 9.1%, while the two bare-hands control accounts each lost 54–56%. The money saved came from exit discipline, not from market selection.
- 2 GLM's discovery is a disaster in the low-price range.** Bench's dual ledger (stop-losses don't pollute accuracy) measured 56.8% overall accuracy (83/146), but <20¢: 1/14 and 20–40¢: 10/31 — low-price long shots were systematically overrated ("12 bets, 0 hits under \$0.30" was the verbatim July–August finding). The 40–80¢ band, at ~70% accuracy, was the only usable range.
- 3 Small positions are uneconomical.** At \$2–5 per position, order rounding, spread and unsellable dust eat any edge a strategy could have.

3. What each stage did

Stage ① — Prehistoric v1 "polymarket-bot" (2026-04-08 ~ 04-16)

The origin of everything: an "Autonomous Polymarket Bot" that drove Gemini Deep Research (and briefly Claude's web UI) through browser automation to research candidate markets and emit structured trade recommendations, targeting mispriced low-probability events. It left behind complete code, a full backup from Apr 15, two dedicated Chrome profiles and a directory full of debug screenshots (the pain of automating web UIs). The "free-of-API-fees, drive-the-web-

version" route was abandoned in favor of v3's paste-the-prompt semi-automation — but the core shape, "AI deep research + structured JSON recommendation", survived from here to the very end. v2 never had its own directory; only a technical document remains (in archive folder 05).

Stage ① — v3 semi-auto initial (2026-04-27, frozen)

Design philosophy: human-machine division of labor. Scanner mechanically filters liquidity/price/depth into a scan report → human takes it to Claude Research for analysis and places limit orders on the website (as maker, saving taker fees) → bot watches every 3 minutes, executing T1/T2/T3 tiered take-profit and W1/W2/W3 reverse stop-loss → a yellow alert prompts a human re-evaluation near targets. Frozen after two commits; the prototype for everything that followed.

Stage ② — Semi-auto main project v4 → v7.5.4 (2026-05-09 ~ 07-24)

The project's main battleground — live money, 132 substantive commits (plus 3,200+ automated cron-backup commits). Human pastes DISCOVERY/REEVAL prompts to Claude and decides; bot scans, monitors, auto-executes take-profit/stop-loss, keeps books, serves the web dashboard. The core assets matured here:

- **Decision engine:** 3-tier stop-loss (convergent/hybrid/event-driven, LLM entry classification), asymmetric trigger pricing (TP uses best_bid against phantom highs, SL uses cur against transient fakes), trailing stops, 2x-sell-all take-profit ladder.
- **Re-evaluation system:** blind-evaluation-first methodology, pre-crash anchor de-biasing, GLM takeover (glm-5.2 + search_pro + max thinking + regex fallback), emergency API pause switch.
- **Infrastructure:** SQLite WAL, dashboard (equity curve / event center / historical position analytics / mobile read-only view), auth & rate limiting, paper trading, DNS-poisoning guard, dual-repo sanitized publishing, cron auto-backup.

On 2026-07-24 the old bot was killed by user decree and its account repurposed as the AUTO project's Bench control account; this directory froze there.

Stage ③ — Fully-automatic AUTO v8.0 → v8.5.5 (2026-07-06 ~ sealed)

Forked 2026-07-06 from semi-auto v7.4.3 as an independent copy (own account/database/port 5052/version line), targeting "fully automatic, zero human". Development protocol: one step at a time with user sign-off, simple hard-coded rules, all new code in standalone new files so future syncs from the old project never conflict. A two-day sprint (07-06~08) opened the whole pipeline:

- **Full pipeline:** scheduled scans 09:00/21:00 (27 whitelisted tags) → streaming GLM discovery (feed-as-you-scan, 4-way concurrency, multi-round agentic search) → hard-coded routing (0.40<price<0.85 real money, else paper) → auto ordering (Kelly sizing + confidence

multiplier + drift re-check) → monitoring → dip-triggered + daily 15:00 GLM re-evaluation → auto exit.

- **New-account battle:** Polymarket's 2026 deposit-wallet architecture requires sig=3 + vendored client 1.0.2 + integer-share rounding — an afternoon of pain; the live-money round trip verified at a total cost of 3 cents.
- **Safety gates:** universe iron rule (only whitelist-tag scanned markets tradable, born of the 07-08 sports mis-buy incident), anti-self-hedge lock, direction-sanity re-ask gate, no-metadata-no-re-eval, 16 fixes from a full code audit.
- **Strategy iteration** (dense mid-late July): event-driven —50% hard stop, convergent/hybrid breach → direct sell, exit-guard removal, q-calibration moved from prompt (×0.5) to Python (×0.8), per-position cap \$15→\$5, monthly drawdown budget removed (see timeline).
- **Three-account experiment** (built 07-23~24): Bench baseline (takes everything, measures GLM's raw accuracy) + Shadow low-price hold (—60% stop derived from a drawdown study of 18 low-price positions), each with its own dashboard.
- **Endgame:** during the 08-12~08-27 outage window Zhipu silently changed its Web Search API (ignores `count`, always returns 50 items) → on resume, context doubled, discovery kept hitting the 20-minute circuit breaker, bills doubled → 09-01 user ordered all paid APIs stopped (v8.5.5), leaving only free price-based monitoring → sealed 09-20.

4. Complete iteration timeline

Stages ①+② (prehistory & semi-auto, 2026-04 ~ 07)

Date	Version	What happened
04-08~16	v1	Prehistoric "polymarket-bot": browser automation drives Gemini Deep Research; route abandoned after the Apr-15 full backup
mid/late Apr	v2	Transitional; no directory survives, only a technical document (archive folder 05)
04-27	v3	Semi-auto initial finalized and frozen: human orders + Claude analysis + tiered TP/SL bot (T1-T3/W1-W3)
05-09	v4.1	Main project baseline first commit; bilingual README, cron auto-backup system
05-10~11	v4.x	Dashboard takes shape: equity curve (Chart.js multi-range), P&L columns, metric cards, confidence persistence
05-19~24	v5.1	Decision-engine rewrite: stop-loss becomes a single 25% rule + 3-tier entry classification (LLM labels convergent/hybrid/event); asymmetric trigger pricing (TP=best_bid / SL=cur, fixing Dell-style false triggers); avg_price convention established

05-24	v5.6	Renumbering; technical report gains the full version history
05-29~30	v5.7	Dashboard auth (Tailscale + password + rate limit); 13 programming-layer bug fixes; private-dev/public-release dual-repo tooling
05-31	v5.8	Holdings tab + bulk re-eval prompt copying; chip state lifecycle (copy = marked)
06-01~03	v5.9~5.10	Cluster dictionary auto-injected into DISCOVERY; three-layer forced-Chinese output; historical positions page + research-grade analytics
06-12	v5.11~5.12	Site polish + Claude JSON fast lane; mobile read-only view /m
06-18~19	v6.0.x	Auto re-evaluation system: concurrency lock + stuck-reset; emergency API pause switch; re-eval throttling (6h baseline + another -10pp triggers); GLM maxed (glm-5.2 + thinking + search_pro) + JSON regex fallback
06-22	v7.0~7.1	Exit-strategy redesign (fixes "selling winners too early" and "round trip to break-even"); two-phase paper trading; full bug audit
07-06~08	v7.4.4~7.4.6	Hybrid tier becomes trailing stop (35% off peak); hybrid breach sells directly without re-eval; entry-price recording bug fixed
07-09~13	v7.5.2~7.5.4	GLM re-eval token budget raised; event-driven positions barred from API stop-loss cancellation; trade-review report + mobile /report page
07-24	—	Old bot terminated (killed by user decree after idling 11 days); its account became AUTO's Bench control; auto-backups end at 12:00 that day

Stage ③ (fully-automatic AUTO, 2026-07 ~ 09)

Date	Version	What happened
07-06	fork	Independent copy forked from semi-auto v7.4.3 (no history/keys/data); isolation iron rules, port 5052
07-07	—	Entry renamed to autobot.py + restart.sh (structural kill-safety: the old docs' <code>pkill main.py</code> can never hit this process)

07-08	—	The longest day: steps 1 (scheduled scan) / 2 (GLM discovery) / 2.5 (daily all-position re-eval sweep) go live; sig=3 + client 1.0.2 cracked, live-money loop verified; steps 3+4 (auto ordering) live = full automation on ; sports mis-buy incident → universe gate; tags-immutable iron rule (dynamic tag feature deleted); <5-candidates-per-tag skip rule; 16-fix full code audit; Claude as GLM-outage fallback only
07-09	—	GLM multi-round agentic search (search→think→search again, up to 8 rounds); 65,536-token output, temperature 0.3; sweep moved 10:00→15:00; streaming pipeline finished (feed-as-you-scan + order-on-arrival)
07-10	—	Reverse recommendation triggers held-position re-eval (reason injected verbatim into prompt); keyword blacklist; sweep 4-way concurrency
07-11	—	Kelly bankroll frozen at \$50 (more cash = more positions, never bigger ones); AUTO technical report + bilingual README
07-12	v8.0.0	AUTO's own version line begins; full-auto monitoring homepage (daily reports / event center / re-eval cards); blacklist → paper-only; confidence multiplier $\times 1.25/1.0/0.75$; Politics tag removed (content filter always blocked); 8.0.1~8.0.4 four rounds of page polish
07-16	v8.0.5	Deposit-adjusted equity curve (deposits no longer break the chart)
07-18	v8.0.6 v8.1.0 v8.2.0 v8.3.0	Monthly drawdown budget gate deleted (\$30 cap was rejecting all new orders); event-driven -50% hard stop + exit guard deleted (GLM's exit executes immediately); q calibration moved to Python, coefficient 0.5→0.8 (trust self 80%); convergent breach → direct sell + intraday re-eval trigger becomes " $\geq 5pp$ retrace from best point" + cancel_autostop removed
07-20~21	v8.3.1~8.3.3	History page folding; 20-minute per-tag hard timeout (a hung GLM call can't freeze the round); prompt-side $\times 0.5$ calibration physically deleted ("whatever raw output is, that's it")
07-23	v8.3.3 v8.4.0	Per-position hard cap \$15→\$5 (user felt single positions too large); Bench baseline strategy (take everything / \$2 / sell at -30% / dual ledger, measuring GLM's raw accuracy); loss post-mortem report
07-24	v8.4.1 v8.5.0~8.5.3	Bench goes live-money (old account's \$90.47 plugged in) + /bench page; Shadow low-price hold (≤ 0.40 / \$2 / -60% stop to resolution, stop value from an 18-position drawdown study); Shadow goes live-money (third account) + /shadow page; churn post-mortem report

08-12~27	—	Outage window (bot not running). Meanwhile Zhipu changed its Web Search API: ignores <code>count</code> , always returns 50 items
08-27	v8.5.4	Resumed; equity curve excludes outage gaps + boot-time phantom points; three-account report generator (asset-truth iron rule)
09-01	v8.5.5	All paid APIs stopped (user decree): root cause = the 50-item search responses pushed context from 150k to 300–500k tokens → 13 circuit-breaker hits in 7 days + prompt-token bills doubled. Discovery key disabled, all re-evaluation off (also fixed the emergency-pause switch not covering the daily sweep); only free price-based monitoring / stop-losses / three dashboards left guarding positions
09-20	sealed	Project declared sealed; all code/data/keys/history archived

5. Final architecture at a glance (plain words)

- **One process, three accounts:** autobot.py (port 5052) runs the main strategy + Bench + Shadow as isolated instances — separate keys/funds/tables, a same-credentials guard prevents cross-wiring.
- **Discovery chain:** scheduled scans over 27 whitelisted tags (medium filter) → tags with ≥ 5 candidates go to GLM (discovery key) → GLM's multi-round search yields JSON recommendations → 0.40~0.85 buys real money (quarter-Kelly on a frozen \$50 reference + confidence multiplier + \$1~5 hard bounds), the rest go to free paper → Bench takes everything, Shadow takes the ≤ 0.40 half.
- **Guarding chain** (pure price logic, zero API, still running at seal): convergent 20% off peak (12% if ≤ 3 days), hybrid 35%, event-driven —50% hard stop + \$0.05 floor, confirmed over consecutive ticks then sold directly; take-profit ladder 2x-sell-all / 0.92-sell-half etc.
- **Re-evaluation chain** (stopped): event-driven ≥ 5 pp-from-best triggers once + daily 15:00 sweep; GLM primary with Claude as outage fallback; direction-sanity re-ask gate against flipped numbers.
- **Pages:** main dashboard (equity curve / daily reports / event center) + /auto candidates + /paper + /history + /bench + /shadow + mobile /m.
- **Safety design:** universe gate (whitelist tag + scan-report double check), blacklist paper-only, anti-self-hedge, 3pp pre-order drift check, DNS-poisoning guard, pending-row-first crash-safe ordering.

6. Incident log (all worth remembering)

Date	Incident / lesson
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07-08	Sports mis-buy: an E2E test injected an out-of-universe sports market (France World Cup) into the pipeline and it bought with real money. User was furious; position closed immediately (a few cents of friction). Birth of the universe-gate iron rule: tests must only use real markets inside the whitelist.
07-08	An afternoon lost to signatures: new account showed \$0 balance + "maker address not allowed" while the private key and address were correct all along — answer: 2026 deposit-wallet architecture requires POLY_SIGNATURE_TYPE=3 + client 1.0.2. Lesson: never apply old-account experience to a new architecture.
07-20	GLM call hung for 2 hours: an Iran summary call froze the whole discovery round while holding the lock. Fix: 20-minute per-tag hard circuit breaker.
07-21	Oversized single position: the formula gave \$11 (~11% of bankroll) to an earthquake market; user ruled the hard cap \$15→\$5.
08-12~27	Upstream silently changed an API: Zhipu Web Search ignores <code>count</code> , always 50 items. Not a single line of our code changed, yet after resume: context doubled, 13 circuit-breaker hits, bills doubled. Lesson: monitor/assert the behavior of upstream APIs — "code unchanged ≠ system unchanged".
08-27	The ledger lies: worthless unsellable positions and vanished positions never enter the per-trade ledger; judging by the ledger alone overstates performance. Established the asset-truth accounting iron rule. Also on record: the old monitor retries selling unsellable positions every 30s forever (can't be fixed in monitor.py per the isolation rule).
ongoing	Local DNS poisoning: *.polymarket.com intermittently fails to resolve on this machine; removing the DNS guard silently paralyzes the whole bot.
ongoing	Zhipu content filter: sensitive tags (Politics, later China) blocked 3/3 rounds; the only cure was removing them from the whitelist.

7. If ever restarted — what to improve (by priority)

1. **Fix the 50-item search problem** (prerequisite for re-enabling APIs): don't wait for upstream — truncate results client-side to 15 items before feeding GLM. One small change, already scoped.
2. **Feed Bench data back into discovery:** the measurements already point the way — below 40¢ accuracy was 11/45, so ban or heavily down-weight that range; the ~70%-accurate 40–80¢ band is the only edible one. The 0.40~0.85 routing line accidentally blocked the worst range, and the Shadow experiment proves "hold cheap long shots to the end" doesn't work.
3. **Calibrate q from data:** AUTO_Q_TRUST=0.8 was a guess. 146 resolved Bench samples are enough to regress a "GLM-reported q vs. realized win rate" calibration curve.

4. **Position-size economics:** at \$2–5, rounding/spread/dust friction dominates. Either scale bankroll so positions are $\geq \$20$, or accept that the cost structure fails at small size.
5. **Upstream API behavior assertions:** alarm on result counts, token usage and per-round latency; auto-pause the money-burning chain on anomaly (this time 13 breaker hits burned for half a month before being noticed).
6. **Dust-position module:** fix the infinite 30s sell-retry in a standalone new file (monitor.py untouchable per the isolation rule), with backoff + forced write-off of dust.
7. **Automatic ledger reconciliation:** compare ledger totals against asset truth daily; red-alert past a threshold instead of discovering the gap in a month-end post-mortem.
8. **Multi-model comparison:** the whole experiment tested only GLM. The Bench framework is ready — plug in a second model (Claude/DeepSeek/...) on the same markets to separate "AI can't do this" from "GLM can't do this".
9. **Notifications / watchdog:** user chose to go without (checks the dashboard personally). For true unattended operation, a dead process with nobody knowing is the hard flaw.
10. **Longer experiment window:** only ~2 months of genuine three-account operation (minus a half-month outage); 146 resolved samples across 5 price buckets is thin.

8. Archive notes

The full archive (including credentials and complete git histories) lives on the owner's local disk. The public GitHub repository is a sanitized snapshot: latest system at top level (`latest_AUTO-v8.5.5_three-account-system/`), all history under `history/` , every secret removed and verified absent. At seal time roughly \$154.58 of assets remained on-chain across the three accounts (the main account still held 27 positions riding to resolution, unattended). The old semi-auto bot must never be restarted — its monitor would sell Bench's positions under the old strategy.